

UNDERSTANDING THE LAW OF LARGE NUMBERS VIA PYTHON

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TRUST THE PROCESS

In Articles No.11-12, we introduced probability distributions and the concepts of expectation and variance. While the expected value provides a measure of the average outcome of a random variable, it is natural to ask whether this theoretical quantity can actually be observed in practice. If we repeatedly perform an experiment, will the average of our observations eventually approach the expected value?

The Law of Large Numbers provides an answer to this question. Casting the mathematical definition aside, it defines the stability and reliability of large sample averages, stating that as the number of observations increases, the sample average converges towards the true expected value. This result forms one of the foundations of probability and statistics, explaining why averages become more stable as more data are collected and why probabilistic models are useful in practice.

THE LAW OF LARGE NUMBERS

What does it mean for a sequence of IID random variables $X_1, X_2, \dots, X_n$ to converge to a random variable X ? There are three different types of convergence:

1. Convergence in distribution
2. Convergence in probability
3. Almost sure convergence

They have different meanings but there are certain implications: almost sure convergence implies convergence in probability which implies convergence in distribution. However, the converse is not true! In essence, convergence in distribution is a statement only about the distributions of the random variables, and doesn't depend on how they are coupled. The sequence of random variables need not even be defined on the same probability space. Convergence in probability is a statement about the limit of a sequence of probabilities. And, almost sure convergence is statement about the probability of a limit.

Let $X_1, X_2, \dots, X_n$ be a sequence of IID random variables with finite expectation $\mathbb{E}[X_i] = \mu$. We say that the sequence converges in distribution to a random variable X if $\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t)$ for every $t \in \mathbb{R}$ at which $F_X(t)$ is continuous. We say that X_n converges to X in probability if for every $\epsilon > 0$, $\lim_{n \rightarrow \infty} P[|X_n - X| > \epsilon] = 0$. And finally, we say that $X_n \rightarrow X$ almost surely if $P[\lim_{n \rightarrow \infty} X_n = X] = 1$.

The **Law of Large Numbers** states that as the sample size n increases, the sample mean $\bar{S}_n = \frac{X_1 + \dots + X_n}{n}$ approaches the true mean μ , that is, $\bar{S}_n \rightarrow \mu$. In words, the average of many independent observations becomes increasingly close to the expected value of the underlying distribution.

There are two versions of the Law of Large numbers, a weak version, and a slightly stronger version. The **Weak Law of Large Numbers** states that the sample mean converges in probability to the expected value $\bar{S}_n \xrightarrow{P} \mu$. The **Strong Law of Large Numbers** states that the sample mean converges almost surely, $\bar{S}_n \xrightarrow{a.s.} \mu$. While the distinction is important mathematically, both results convey the same practical message: given sufficiently many observations, the sample average will become arbitrarily close to the expected value.

Consider repeatedly tossing a fair coin. Let $X = \begin{cases} 1, & \text{Heads} \\ 0, & \text{Tails} \end{cases}$. Thus, the true mean is $\mathbb{E}[X] = 0.5$. After a small number of tosses, we obtain the sample $n = 3$; $[1, 0, 0]$, i.e., 1 heads and 2 tails, the proportion of heads is $\frac{1}{3}$. However, as the number of tosses increases, the proportion of heads tends to move closer and closer to 0.5. For example, the new sample after $n = 10$; $[1, 0, 0, 1, 1, 0, 1, 1, 1, 0]$, i.e., 7 heads and 4 tails, thus, the proportion is 0.6. And after 10,000 tosses, the proportion of heads is likely to be much closer to 0.5.

UTILISING PYTHON

Simulating sample sizes large enough to illustrate the Law of Large Numbers can quickly become impractical by hand. Fortunately, code allows us to generate and analyse numbers of observations with ease. Thus, to help build an intuitive understanding, I have written simple code in Python that helps visualise the Law of Large Numbers.

Consider a standard normal distribution, we can investigate convergence in probability by examining the behaviour of the sample mean as the sample size increases. For $n = 100$, the generated output is 0.932.

```

1| #Import module
2| import numpy as np
3|
4| epsilon = 0.01 # Set tolerance
5| l = [] # Create empty array
6| for i in range(1000): # Repeat experiment 1000 times
7|     n = np.random.normal(0, 1, 100) # Generate a sample of 100 values from a standard normal distribution
8|     xbarn = np.mean(n) # Calculate sample mean
9|     if np.abs(xbarn) > epsilon: # Check whether sample mean differs from expected value by more than tolerance
10|         l.append(1)
11|     else:
12|         l.append(0)
13| print(sum(l)/len(l)) # Compute probability

```

Increasing the sample size to $n = 10,000$ reduces this value significantly to 0.303.

```

1| import numpy as np
2|
3| epsilon = 0.01
4| l = []
5| for i in range(1000):
6|     n = np.random.normal(0, 1, 10000)
7|     xbarn = np.mean(n)
8|     if np.abs(xbarn) > epsilon:
9|         l.append(1)
10|    else:
11|        l.append(0)
12| print(sum(l)/len(l))

```

Finally, when $n = 1 \times 10^6$, the probability falls to 0.

```

1| import numpy as np
2|
3| epsilon = 0.01
4| l = []
5| for i in range(1000):
6|     n = np.random.normal(0, 1, 1000000)
7|     xbarn = np.mean(n)
8|     if np.abs(xbarn) > epsilon:
9|         l.append(1)
10|    else:
11|        l.append(0)
12| print(sum(l)/len(l))

```

As the sample size grows, the probability of observing a large deviation from the true mean decreases, illustrating convergence in probability and providing evidence for the Weak Law of Large Numbers.

The Law of Large Numbers is particularly important in Monte Carlo simulations, where expected values are approximated using a large number of randomly generated scenarios. As the number of simulations increases, the estimated value converges towards the true expectation. Similarly, the Law of Large Numbers helps explain why long-term average returns, default rates, *etc.* become more stable when calculated from large datasets.

CONCLUSION

The Law of Large Numbers provides a bridge between theoretical probability and observed data. By showing that sample averages converge to expected values, it helps to explain why probabilistic models are useful and why estimates based on large samples are often reliable.

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